

# Lin Xin

## GAME PROGRAMMER

Singapore | soniclinxin@gmail.com | <https://linx-z0.github.io/portfolio/> | <https://www.linkedin.com/in/xinl0/>

## PROFESSIONAL SUMMARY

Computer Science and Game Design student at DigiPen Institute of Technology Singapore with a focus on gameplay programming, C++ and game systems, supported by a strong foundation in mathematics and computer science. Experienced in developing games with custom and commercial engines, implementing gameplay systems, debugging, and working in multidisciplinary teams. Proficient in C, C# and C++, with knowledge of data structures, algorithms and mathematical concepts relevant to game development.

## SKILLS

**Programming:** C | C# | C++

**Engine & Tools:** Unity | Unreal Engine | GitHub | Visual Studio | Figma | Jira | Notion | Canva

**Technical:** Probability & Statistics | Linear Algebra | Calculus | Data Structures | Algorithms | Algorithm Analysis | Math Library Design | Problem Solving

**Technical Design:** Gameplay Implementation | Gameplay Tuning | AI Behavior Design | Level Implementation | System Integration | Debugging | Technical Documentation

**Game Design:** Game Systems Design | Mechanics Design | Gameplay Loops | Combat Design | Encounter Design | Level Design | Player Progression | Prototyping | Game Balancing | Playtesting | Iterative Design | Pacing | Game Design Documentation (GDD)

## GAME PROJECTS

### Death's Refrain, 2D Top-Down Grid-Based Game – Design Lead

*DigiPen Institute of Technology Singapore | 9-Person Team | Custom Game Engine | Year 2 Trimester 1 & 2*

- Developed a 2D top-down, grid-based tactical game as part of a six-programmer team using a custom-built game engine following an initial Unity prototype.
- Created a tactical combat system built around Action Point management, grid movement, melee and ranged combat, defensive mechanics and turn-based decision-making.
- Worked within a combat design that emphasized positioning, line-of-sight, resource management and enemy prioritization.
- Contributed to encounters featuring distinct enemy archetypes including melee pursuit, ranged line-of-sight attacks, respawning enemies and a multi-attack boss encounter.
- Helped develop a progression structure incorporating increasingly complex enemy compositions, environmental hazards, pickups, permanent upgrades and boss encounters.
- Lead the integration and iteration of gameplay, level progression, UI feedback, audiovisual elements and narrative sequences.
- Worked from a detailed Game Design Document and gameplay wiki covering mechanics, enemies, progression, controls, levels, UI and narrative systems.

### Adaptive Alien AI Simulation – Gameplay Programmer

*DigiPen Institute of Technology Europe, Bilbao | 4-Person Team | C++ | Technical Design | Unreal Engine | Year 2 Trimester 3*

- Developed an adaptive enemy AI system in C++ within Unreal Engine for a top-down Simulation inspired by Alien: Isolation
- Designed a reusable Smart Interactable system supporting objects such as cabinets and vents, including reservation, occupancy and interaction-point logic.
- Implemented gameplay architecture separating C++ AI systems from Blueprint-based environments, interactables and test NPCs.
- Developed and tested AI navigation, hiding interactions and behavioral responses within a NavMesh-based environment.
- Diagnosed and resolved issues involving AI possession, navigation, collision and Blueprint/C++ integration.

### Overseas, 2D Top-Down Fishing Survival game– Gameplay Programmer

*DigiPen Institute of Technology Singapore | 5-Person Team | C++ | Technical Design | Year 1 Trimester 2*

- Developed gameplay systems for a fishing/survival game, with primary contributions to fishing mechanics, environment interactions, resource systems, upgrades, and gameplay HUD elements

- Built the fishing gameplay framework, including fishing-spot spawning and collision, catch-driven spot replacement, fishing states, quick time event (QTE) input handling, and a rarity-aware QTE interface.
- Implemented the health and supply gameplay loop and HUD, including passive supply depletion, health regeneration/penalties based on supply level, upgrade-modified values, and dynamic sprite-based UI feedback.
- Developed a shipwreck resource system supporting randomized spawning, collision detection, supply rewards, removal/respawning, and a maintenance of a fixed number of world objects.

## PROFESSIONAL EXPERIENCE

### DigiPen Institute of Technology Singapore Teaching Assistant

Sep 2025 - Jan 2026

- Assisted in teaching mathematics and computer environment during scheduled classes.
- Guided students through mathematical problems, technical concepts and engineering exercises.
- Conducted additional after-school consultation and support sessions for students requiring further help.
- Adapted explanations and problem-solving approaches based on individual student understanding.
- Developed strong technical communication, mentoring, and problem-solving skills through regular interaction with students.

### Singapore Cybersports & Online Gaming Association - SCOGA Events & Media Intern

Apr 2019 - Oct 2019

- Contributed to the planning and execution of gaming and esports events including Campus Game Fest.
- Created and managed social media content supporting events and community initiatives.
- Assisted in the production of WELP Tonight — Welcome Everyone, Let's Play, a Twitch programme featuring celebrity guests and gaming culture.
- Supported event operations and media production within Singapore's gaming community.

## EDUCATION

### DigiPen Institute of Technology Singapore, Singapore BSc Computer Science in Interactive Media and Game Development

Sep 2024 - Present

- Developing games across 2D, 3D, VR, and custom-engine environments with experience spanning both game design and technical implementation.

### Eunoia Junior College GCE A-Level

Jan 2020 - Dec 2021

## LANGUAGES & CROSS-CULTURAL EXPERIENCE

English: Fluent | Chinese: Professional Working Proficiency

Bicultural Studies Programme (BSP): Completed a four-year MOE programme from Secondary 3 to JC 2 focused on contemporary China, the West, and their intersections from a Singaporean perspective.